



Lessons



Using Systems

A system is a group of related elements that work together for a common goal. An airport has a system of runways, as well as an organized network of computers and air traffic controllers to keep travelers safe. The human body has many systems—the circulatory system moves blood throughout the body, while the digestive system takes in food and nutrients and eliminates waste.

In mathematics, equations and inequalities form systems. Systems have more than one variable. In this lesson, you will learn how to write and solve systems with two variables: x and y .

The lesson presents multiple ways of solving systems. Take the time to understand each method. Each way becomes another tool in your algebra toolbox. Good test takers know that systems are the key to solving complex problems simply.

Word problems tell a story. The words give you the facts you need to solve the problem. The words also show how the facts are related. In an algebra problem, one or more of the numbers are unknown.

For some problems, organizing the information in a chart before writing an equation is helpful. Study the example below to see how to use a chart to write an equation.

EXAMPLE

Hide

Chelsea, Robin, and Dan start a dog obedience school. Because they will not be investing equal amounts of time and money, they will not split the profits evenly. They agree that Robin will receive twice as much as Dan, and Chelsea will receive twice as much as Robin. How much is Robin’s share of \$3,500?

Step 1

Create a chart showing what you know. You know the least about Dan’s share. Let x represent his share. Fill out the rest of the chart using information from the problem.

Dan’s Share	Robin’s Share (2 times Dan’s share)	Chelsea’s Share (2 times Robin’s share)	Total
x	$2x$	$2(2x)$	\$3,500

Step 2

Write an equation based on the chart.

$x + 2x + 2(2x) = \$3,500$

Step 3

Solve for x .

$$\begin{aligned} x + 2x + 4x &= \$3,500 \\ 7x &= \$3,500 \\ x &= \$500 \end{aligned}$$

Step 4

Answer the question asked in the problem. You need to find Robin’s share. Go back to the chart. Robin’s share is represented by $2x$.

Substitute \$500 for x : $2(\$500) = \$1,000$

Robin’s share of \$3,500 is \$1,000.

Check to see whether your answer makes sense. Chelsea earned twice Robin’s share, so Chelsea must have earned \$2,000. Does the total equal \$3,500?

$\$500 + \$1,000 + \$2,000 = \$3,500$ The answer makes sense.

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Category

Lesson 4: Systems of Equations

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